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ADVANCED OPTIMIZATION



You already know optimization. In Chapter 3, we covered gradient ascent/descent, which lets us “climb hills” to find a maximum or minimum.

Any optimization problem can be thought of as a version of hill climbing: we strive to find the best possible outcome out of a huge range of possibilities. The gradient ascent tool is simple and elegant, but it has an Achilles’ heel: it can lead us to find a peak that is only locally optimal, not globally optimal. In the hill-climbing analogy, it might take us to the top of a foothill, when going downhill for just a little while would enable us to start scaling the huge mountain that we actually want to climb. Dealing with this issue is the most difficult and crucial aspect of advanced optimization.